

Instrumental Variables Regression

Quantitative Economics

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Based on Stock & Watson, *Introduction to Econometrics*, Chapter 12.

Roadmap

1. Motivation: Why Do We Need IV?
2. Exogeneity: A Deeper Look
3. Sources of Endogeneity: Theory and Simulation
4. The Two Stage Least Squares (TSLS) Estimator
5. Why IV Works: Consistency of TSLS
6. Sampling Distribution of the TSLS Estimator
7. Illustrative Examples
8. The General IV Regression Model
9. Checking Instrument Validity
10. Application: Cigarette Demand with Panel Data
11. Where Do Valid Instruments Come From?
12. Advanced Topics
13. Summary

The fundamental problem: endogeneity

Consider the regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- OLS consistency requires $\text{cov}(X_i, u_i) = 0$.
- When this fails, OLS is **inconsistent**: $\hat{\beta}_1^{OLS} \xrightarrow{p} \beta_1 + \frac{\text{cov}(X_i, u_i)}{\text{var}(X_i)} \neq \beta_1$.
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Three classical sources of endogeneity

1. **Omitted variables**: A variable in u_i is correlated with X_i .
2. **Measurement error**: X_i is measured with noise; the noise enters u_i .
3. **Simultaneous causality**: $X \rightarrow Y$ and $Y \rightarrow X$ simultaneously.

Endogeneity vs. exogeneity – terminology

Definitions

- A variable is **endogenous** if $\text{corr}(X_i, u_i) \neq 0$.
Historically: “determined *within* the model.”
- A variable is **exogenous** if $\text{corr}(X_i, u_i) = 0$.
Historically: “determined *outside* the model.”

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Intuition

In a supply–demand system, both price and quantity are determined *within* the model (by the intersection of the curves). Both are endogenous. An outside shock – say rainfall affecting crop yields – is determined *outside* the model and is exogenous.

The key idea behind IV regression

Think of the variation in X as having **two components**:

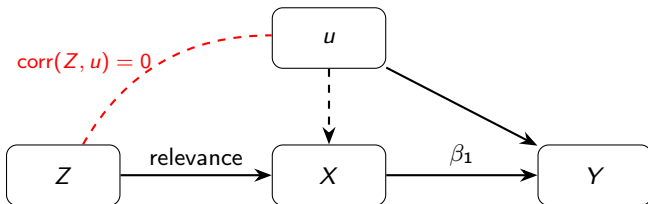
1. A **problematic** component correlated with u
(this causes bias).
2. A **problem-free** component uncorrelated with u
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2. A **problem-free** component uncorrelated with u (this is what we want to use).

Strategy: Find an additional variable Z – an **instrumental variable** – that captures the problem-free variation in X .



Two conditions for a valid instrument

An instrumental variable Z must satisfy:

Condition 1: Instrument Relevance

$$\text{corr}(Z_i, X_i) \neq 0$$

Z must be correlated with the endogenous regressor X .

Condition 2: Instrument Exogeneity

$$\text{corr}(Z_i, u_i) = 0$$

Z must be uncorrelated with the error term u .

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Why both are needed

- Relevance without exogeneity $\Rightarrow$ IV captures contaminated variation $\Rightarrow$ inconsistent.
- Exogeneity without relevance $\Rightarrow Z$ carries no information about $X \Rightarrow$ division by zero; estimator undefined.

Three layers of exogeneity

Textbooks state instrument exogeneity as $\text{corr}(Z_i, u_i) = 0$. But there are three progressively stronger conditions:

Level 1: Zero Correlation (Weakest)

$$\text{cov}(Z_i, u_i) = 0$$

The *minimum* for TSLS consistency. Z and u are linearly unrelated.

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Level 3: Full Independence (Strongest)

$$u_i \perp\!\!\!\perp Z_i$$

The *entire distribution* of u is unaffected by Z . This is what “as good as randomly assigned” really means.

Hierarchy: mean independence $\Rightarrow$ zero correlation

Proof: $E(u_i | Z_i) = 0 \implies \text{cov}(Z_i, u_i) = 0$

By the *law of iterated expectations*:

$$E(Z_i u_i) = E[E(Z_i u_i | Z_i)] = E[Z_i \cdot \underbrace{E(u_i | Z_i)}_{=0}] = 0$$

$$E(u_i) = E[\underbrace{E(u_i | Z_i)}_{=0}] = 0$$

Therefore:

$$\text{cov}(Z_i, u_i) = E(Z_i u_i) - E(Z_i)E(u_i) = 0 - 0 = 0 \quad \square$$

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The converse is FALSE

$$\text{cov}(Z_i, u_i) = 0 \not\Rightarrow E(u_i | Z_i) = 0.$$

Counterexample: Let $Z \sim N(0, 1)$ and $u = Z^2 - 1$.

- $E(u) = 0$ and $\text{cov}(Z, u) = E(Z^3) - E(Z)E(Z^2 - 1) = 0$. ✓

Why the distinction matters

1. **Zero correlation** is enough for **consistency** of TSLS.
2. **Mean independence** additionally ensures that standard errors, t -statistics, and confidence intervals have their usual large-sample properties.
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Practical implication

When researchers argue for instrument exogeneity, they typically need arguments strong enough for *at least* mean independence. A variable that satisfies only zero correlation but has nonlinear dependence with u may produce misleading inference even if the point estimate is consistent.

What happens when exogeneity fails?

If $\text{cov}(Z_i, u_i) \neq 0$, the TSLS estimator converges to the **wrong number**:

$$\hat{\beta}_1^{TSLS} \xrightarrow{p} \beta_1 + \frac{\text{cov}(Z_i, u_i)}{\text{cov}(Z_i, X_i)}$$

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- The **asymptotic bias** is $\frac{\text{cov}(Z_i, u_i)}{\text{cov}(Z_i, X_i)}$.
- Compare with OLS bias: $\frac{\text{cov}(X_i, u_i)}{\text{var}(X_i)}$.

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Key danger

Even a *small* violation of exogeneity produces a *large* bias if the instrument is weak ($\text{cov}(Z, X)$ is small). Weak instruments **amplify** exogeneity violations.

See Python notebook: `lectures/code/01_exogeneity_deep_dive.ipynb` for Monte Carlo simulations.

How practitioners assess exogeneity

1. **Economic reasoning** (always required)
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3. **J -test** (when overidentified: $m > k$)
 - Tests whether “extra” instruments are exogenous.
 - $J = mF \sim \chi^2_{m-k}$ under H_0 : all instruments exogenous.
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Bottom line

Exogeneity is ultimately an **assumption**, not something fully verifiable from data. The credibility of IV analysis rests on the persuasiveness of the exogeneity argument.

Roadmap for this section

We now study each source of endogeneity in detail:

1. **Omitted Variable Bias (OVB)** – a confounder in the error term
2. **Measurement Error (Errors-in-Variables)** – noise creates a mechanical correlation
3. **Simultaneous Causality** – X and Y are jointly determined

For each, we will:

- Derive the bias formula mathematically
- Show precisely *how* $\text{cov}(X_i, u_i) \neq 0$ arises
- Determine the direction and magnitude of the bias
- Explain how IV resolves the problem

Accompanying simulations: `lectures/code/02_sources_of_endogeneity.ipynb`

Source 1: Omitted Variable Bias – setup

True data generating process:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \varepsilon_i, \quad E(\varepsilon_i \mid X_i, W_i) = 0$$

Estimated model (we cannot observe W_i):

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When does endogeneity arise?

$$\text{cov}(X_i, u_i) = \text{cov}(X_i, \beta_2 W_i + \varepsilon_i) = \beta_2 \cdot \text{cov}(X_i, W_i) + \underbrace{\text{cov}(X_i, \varepsilon_i)}_{=0}$$

So $\text{cov}(X_i, u_i) \neq 0$ if and only if **both**:

1. $\beta_2 \neq 0$: the omitted variable actually affects Y
2. $\text{cov}(X_i, W_i) \neq 0$: the omitted variable is correlated with X

Omitted Variable Bias – the formula

The OLS probability limit when W is omitted:

$$\hat{\beta}_1^{OLS} \xrightarrow{p} \beta_1 + \underbrace{\beta_2 \cdot \frac{\text{cov}(X_i, W_i)}{\text{var}(X_i)}}_{\text{Omitted Variable Bias}}$$

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Direction of bias

	$\text{cov}(X, W) > 0$	$\text{cov}(X, W) < 0$
$\beta_2 > 0$	Positive bias $\uparrow$	Negative bias $\downarrow$
$\beta_2 < 0$	Negative bias $\downarrow$	Positive bias $\uparrow$

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Example

Estimating the return to education (X), omitting ability (W). If $\beta_2 > 0$ (ability raises wages) and $\text{cov}(\text{education}, \text{ability}) > 0$ (smarter people get more education), then OLS **overestimates** the return to education.

OVB – IV solution

Problem: We cannot include W because it is unobserved.

Solution: Find an instrument Z such that:

- $\text{cov}(Z_i, X_i) \neq 0$ (relevance)
- $\text{cov}(Z_i, u_i) = \beta_2 \text{cov}(Z_i, W_i) + \text{cov}(Z_i, \varepsilon_i) = 0$ (exogeneity)

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$$\text{cov}(Z_i, W_i) = 0 \quad (\text{and } \text{cov}(Z_i, \varepsilon_i) = 0)$$

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Example (Angrist & Krueger, 1991)

Return to education (X), omitting ability (W).

Instrument: *quarter of birth* (Z) – affects years of schooling through compulsory schooling laws, but is plausibly unrelated to innate ability.

Source 2: Measurement Error – setup

True model: $Y_i = \beta_0 + \beta_1 X_i^* + \varepsilon_i$, $E(\varepsilon_i | X_i^*) = 0$
where X_i^* is the *true* value (unobserved). We observe:

$$X_i = X_i^* + w_i$$

with **classical measurement error**: $E(w_i) = 0$, $\text{cov}(w_i, X_i^*) = 0$,
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Substitute $X_i^* = X_i - w_i$ into the true model:

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The endogeneity

$$\begin{aligned}\text{cov}(X_i, u_i) &= \text{cov}(X_i^* + w_i, \varepsilon_i - \beta_1 w_i) \\ &= \underbrace{\text{cov}(X_i^*, \varepsilon_i)}_0 - \beta_1 \underbrace{\text{cov}(X_i^*, w_i)}_0 + \underbrace{\text{cov}(w_i, \varepsilon_i)}_0 - \beta_1 \underbrace{\text{var}(w_i)}_{\sigma_w^2} \\ &= -\beta_1 \sigma_w^2 \neq 0\end{aligned}$$

Measurement Error – attenuation bias

The OLS probability limit:

$$\begin{aligned}\hat{\beta}_1^{OLS} &\xrightarrow{p} \beta_1 + \frac{\text{cov}(X_i, u_i)}{\text{var}(X_i)} = \beta_1 + \frac{-\beta_1 \sigma_w^2}{\sigma_{X^*}^2 + \sigma_w^2} \\ &= \beta_1 \left(\frac{\sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_w^2} \right)\end{aligned}$$

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Since $\frac{\sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_w^2} < 1$:

Attenuation bias

OLS with classical measurement error is **biased toward zero**. The coefficient is “attenuated.”

- If $\beta_1 > 0$: OLS underestimates the true effect.
- If $\beta_1 < 0$: OLS overestimates (closer to zero) the true effect.
- More noise ($\sigma_w^2 \uparrow$) $\Rightarrow$ more attenuation.

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The ratio $\lambda = \sigma_{X^*}^2 / (\sigma_{X^*}^2 + \sigma_w^2)$ is called the **reliability ratio** or **signal-to-total-variance ratio**.

Measurement Error – IV solution

We need an instrument Z such that:

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What instruments work?

- A *second independent measurement* of X^* (e.g., a different survey question measuring the same concept)
- Any variable that predicts X^* but is unrelated to the measurement noise w and the structural error ε

Source 3: Simultaneous Causality – setup

System of simultaneous equations (supply and demand):

$$\text{Demand : } Q_i = \alpha_0 + \alpha_1 P_i + u_i^d \quad (\alpha_1 < 0)$$

$$\text{Supply : } Q_i = \gamma_0 + \gamma_1 P_i + u_i^s \quad (\gamma_1 > 0)$$

Equilibrium: $Q_i^d = Q_i^s$ simultaneously determines both P_i and Q_i .

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Key observation

The equilibrium price P_i depends on the demand shock u_i^d . So $\text{cov}(P_i, u_i^d) \neq 0$: the regressor in the demand equation is **endogenous**.

Simultaneous Causality – deriving the bias

(1/3)

Goal: Estimate the demand elasticity α_1 from $Q_i = \alpha_0 + \alpha_1 P_i + u_i^d$.

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We need two ingredients from the equilibrium price $P_i = \underbrace{\frac{\gamma_0 - \alpha_0}{\alpha_1 - \gamma_1}}_{\bar{P} \text{ (constant)}} + \frac{u_i^s - u_i^d}{\alpha_1 - \gamma_1}$:

1. $\text{cov}(P_i, u_i^d)$
2. $\text{var}(P_i)$

Simultaneous Causality – deriving the bias (1/3)

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Recall the OLS probability limit formula:

$$\hat{\alpha}_1^{OLS} \xrightarrow{p} \alpha_1 + \frac{\text{cov}(P_i, u_i^d)}{\text{var}(P_i)}$$

We need two ingredients from the equilibrium price $P_i = \underbrace{\frac{\gamma_0 - \alpha_0}{\alpha_1 - \gamma_1}}_{\bar{P} \text{ (constant)}} + \frac{u_i^s - u_i^d}{\alpha_1 - \gamma_1}$:

1. $\text{cov}(P_i, u_i^d)$
2. $\text{var}(P_i)$

Assumptions: $u^d \perp u^s$, both mean-zero with variances σ_d^2 and σ_s^2 .

Simultaneous Causality – deriving the bias (2/3)

Step 1: Compute $\text{cov}(P_i, u_i^d)$.

Since $\bar{P}$ is a constant, only the stochastic part matters:

$$\text{cov}(P_i, u_i^d) = \text{cov}\left(\frac{u_i^s - u_i^d}{\alpha_1 - \gamma_1}, u_i^d\right) = \frac{1}{\alpha_1 - \gamma_1} \left[\underbrace{\text{cov}(u_i^s, u_i^d)}_{=0} - \text{var}(u_i^d) \right] = \frac{-\sigma_d^2}{\alpha_1 - \gamma_1}$$

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Since $\alpha_1 < 0 < \gamma_1$, we have $\alpha_1 - \gamma_1 < 0$, so $\text{cov}(P_i, u_i^d) > 0$. **Positive demand shocks raise the equilibrium price.**

Simultaneous Causality – deriving the bias

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Step 2: Compute $\text{var}(P_i)$.

$$\text{var}(P_i) = \text{var}\left(\frac{u_i^s - u_i^d}{\alpha_1 - \gamma_1}\right) = \frac{1}{(\alpha_1 - \gamma_1)^2} \left[\sigma_s^2 + \sigma_d^2 \right]$$

(using $u^d \perp u^s$ so the cross-covariance vanishes)

Simultaneous Causality – deriving the bias

(3/3)

Step 3: Substitute into the probability limit.

$$\hat{\alpha}_1^{OLS} \xrightarrow{p} \alpha_1 + \frac{\text{cov}(P_i, u_i^d)}{\text{var}(P_i)} = \alpha_1 + \frac{\frac{-\sigma_d^2}{\alpha_1 - \gamma_1}}{\frac{\sigma_s^2 + \sigma_d^2}{(\alpha_1 - \gamma_1)^2}}$$

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Simplify:

$$= \alpha_1 + \frac{-\sigma_d^2}{\alpha_1 - \gamma_1} \cdot \frac{(\alpha_1 - \gamma_1)^2}{\sigma_s^2 + \sigma_d^2} = \alpha_1 - \frac{\sigma_d^2(\alpha_1 - \gamma_1)}{\sigma_s^2 + \sigma_d^2}$$

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Combine into a single fraction:

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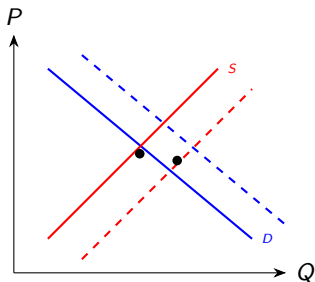
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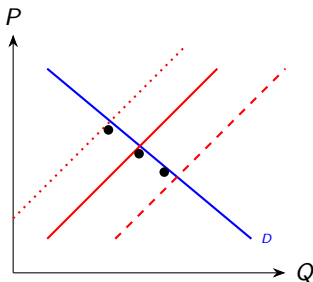
OLS probability limit

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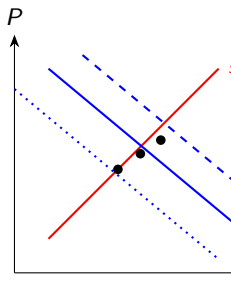
Wright's insight: shifting one curve to identify the other



(a) Both shift



(b) Only supply shifts
 $\Rightarrow$ traces demand



(c) Only demand shifts
 $\Rightarrow$ traces supply

IV idea: Find a variable Z that shifts *only supply* (a **supply shifter**). Then variation in equilibrium (P, Q) traces out the demand curve $\Rightarrow$ we can estimate α_1 .

Simultaneous Causality – IV solution

To estimate the **demand** elasticity, we need a **supply shifter** Z :

- Include Z in the supply equation: $Q_i = \gamma_0 + \gamma_1 P_i + \delta Z_i + u_i^s$
- Z is relevant: $\text{cov}(Z_i, P_i) \neq 0$ (shifting supply changes equilibrium price)
- Z is exogenous: $\text{cov}(Z_i, u_i^d) = 0$ (Z does not directly shift demand)

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Examples of supply shifters for demand estimation

- Philip Wright: *rainfall* in dairy regions $\rightarrow$ butter supply
- Cigarette taxes: *excise taxes* shift supply (increase after-tax price)
- Oil market: *geopolitical disruptions* shift oil supply

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Examples of supply shifters for demand estimation

- Philip Wright: *rainfall* in dairy regions $\rightarrow$ butter supply
- Cigarette taxes: *excise taxes* shift supply (increase after-tax price)
- Oil market: *geopolitical disruptions* shift oil supply

To estimate the **supply** elasticity, reverse the logic: use a **demand shifter** as the instrument (e.g., income shocks, population changes).

Summary: three sources, one unified framework

Source	Why $\text{cov}(X, u) \neq 0$	OLS bias direction	IV requirement
Omitted variables	u contains W ; W correlated with X	Depends on signs of β_2 , $\text{cov}(X, W)$	Z uncorrelated with W
Measurement error	w enters both X and u	Toward zero (attenuation)	Z uncorrelated with w
Simultaneity	X determined by equilibrium involving u	Toward "other equation" slope	Z shifts other equation only

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The common thread

All three create $\text{cov}(X_i, u_i) \neq 0$. IV provides a **unified solution**: find a variable Z that isolates exogenous variation in X .

TSLS: the big picture

Goal: Isolate the exogenous variation in X and use only that to estimate β_1 .

Two stages:

1. **First stage:** Regress X on Z to extract the “clean” component.
2. **Second stage:** Regress Y on the predicted $\hat{X}$ from stage 1.

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1. **First stage:** Regress X on Z to extract the “clean” component.
2. **Second stage:** Regress Y on the predicted $\hat{X}$ from stage 1.

Intuition

The first stage *purifies* X by projecting it onto the instrument Z . The predicted value $\hat{X}$ inherits Z 's exogeneity. The second stage uses only this purified variation.

First stage: decomposing X

The population first-stage regression:

$$X_i = \pi_0 + \pi_1 Z_i + v_i \quad (12.2)$$

- $\pi_0 + \pi_1 Z_i$: the part of X_i that can be predicted by Z_i .
Because Z_i is exogenous, this component is **uncorrelated with u_i** .
- v_i : the “remainder” – this is the **problematic** part correlated with u_i .

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- v_i : the “remainder” – this is the **problematic** part correlated with u_i .

In practice, π_0 and π_1 are unknown. We estimate them by OLS:

$$\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i$$

$\hat{X}_i$ is the **fitted value** from the first stage.

Second stage: using the purified $\hat{X}$

Second stage: Regress Y_i on $\hat{X}_i$ using OLS.

$$Y_i = \beta_0 + \beta_1 \hat{X}_i + \text{error}_i$$

The OLS estimators from this second-stage regression are the TSLS estimators $\hat{\beta}_0^{TSLS}$ and $\hat{\beta}_1^{TSLS}$.

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Important practical note

In practice, both stages are performed automatically by econometric software. The standard errors from a “manual” second-stage OLS regression are **incorrect** because they do not account for the estimation uncertainty from the first stage.

The TSLS formula: $\hat{\beta}_1^{TSLS} = s_{ZY}/s_{ZX}$

Theorem (TSLS Formula)

With a single regressor X and a single instrument Z :

$$\hat{\beta}_1^{TSLS} = \frac{s_{ZY}}{s_{ZX}}$$

where s_{ZY} is the sample covariance between Z and Y , and s_{ZX} is the sample covariance between Z and X .

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Interpretation: The TSLS estimator is the ratio of the reduced-form effect ($Z \rightarrow Y$) to the first-stage effect ($Z \rightarrow X$). This is sometimes called the **Wald estimator**.

Deriving the TSLS formula (1/3)

Starting point: TSLS is OLS of Y on $\hat{X}$, so by the OLS slope formula:

$$\hat{\beta}_1^{TSLS} = \frac{s_{\hat{X}Y}}{s_{\hat{X}}^2}$$

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Key substitution: The first stage gives $\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i$. Its sample mean is:

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Therefore the deviation from the mean simplifies to:

$$\hat{X}_i - \bar{\hat{X}} = \hat{\pi}_1 (Z_i - \bar{Z})$$

Intuition: All variation in $\hat{X}$ comes from Z through the first-stage slope $\hat{\pi}_1$. The intercept $\hat{\pi}_0$ cancels out.

Deriving the TSLS formula (2/3)

Substitute $\hat{X}_i - \bar{\hat{X}} = \hat{\pi}_1(Z_i - \bar{Z})$ into each piece.

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Numerator – sample covariance of $\hat{X}$ and Y :

$$s_{\hat{X}Y} = \frac{1}{n-1} \sum_{i=1}^n \hat{\pi}_1(Z_i - \bar{Z})(Y_i - \bar{Y}) = \hat{\pi}_1 \underbrace{\frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})(Y_i - \bar{Y})}_{= s_{ZY}} = \hat{\pi}_1 s_{ZY}$$

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Ratio:

$$\hat{\beta}_1^{TSLS} = \frac{s_{\hat{X}Y}}{s_{\hat{X}}^2} = \frac{\hat{\pi}_1 s_{ZY}}{\hat{\pi}_1^2 s_Z^2} = \frac{s_{ZY}}{\hat{\pi}_1 s_Z^2}$$

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Interpretation

The TSLS estimator is the ratio of:

- **Reduced form** s_{ZY} : how much Y moves when Z moves (total effect of Z on Y through X)
- **First stage** s_{ZX} : how much X moves when Z moves

This is the **Wald estimator**: $\hat{\beta}_1^{TSLS} = \frac{\text{reduced form}}{\text{first stage}}.$

Consistency of TSLS: the key insight

Starting from $Y_i = \beta_0 + \beta_1 X_i + u_i$, take covariances with Z_i :

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Under **instrument exogeneity**: $\text{cov}(Z_i, u_i) = 0$.

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Under **instrument exogeneity**: $\text{cov}(Z_i, u_i) = 0$.

Under **instrument relevance**: $\text{cov}(Z_i, X_i) \neq 0$.

Solving:

$$\boxed{\beta_1 = \frac{\text{cov}(Z_i, Y_i)}{\text{cov}(Z_i, X_i)}}\tag{12.6}$$

This is the **population** analog of the TSLS formula.

From population to sample: consistency

The sample covariance is a consistent estimator of the population covariance:

$$s_{ZY} \xrightarrow{P} \text{cov}(Z_i, Y_i), \quad s_{ZX} \xrightarrow{P} \text{cov}(Z_i, X_i)$$

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Therefore, by the continuous mapping theorem (Slutsky):

$$\widehat{\beta}_1^{TSLs} = \frac{s_{ZY}}{s_{ZX}} \xrightarrow{p} \frac{\text{cov}(Z_i, Y_i)}{\text{cov}(Z_i, X_i)} = \beta_1 \quad (12.7)$$

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Key takeaway

The TSLS estimator is **consistent** for β_1 under the two conditions for instrument validity. This holds regardless of the correlation between X and u – the endogeneity that plagues OLS is completely circumvented.

Expressing TSLS in terms of errors

From $Y_i - \bar{Y} = \beta_1(X_i - \bar{X}) + (u_i - \bar{u})$, substitute into s_{ZY} :

$$\begin{aligned}s_{ZY} &= \frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})(Y_i - \bar{Y}) \\ &= \beta_1 s_{ZX} + \frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})u_i\end{aligned}$$

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where the last step uses $\sum (Z_i - \bar{Z})\bar{u} = 0$.

Dividing by s_{ZX} and multiplying numerator and denominator by $(n-1)/n$:

$$\boxed{\hat{\beta}_1^{TSLS} = \beta_1 + \frac{\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z})u_i}{\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z})(X_i - \bar{X})}} \quad (12.20)$$

This is the fundamental equation for analyzing the distribution of TSLS.

Asymptotic normality of TSLS (Appendix 12.3)

From Eq. (12.20), focus on the **numerator**:

$$\bar{q} = \frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z}) u_i \approx \frac{1}{n} \sum_{i=1}^n \underbrace{(Z_i - \mu_Z)}_{q_i} u_i$$

Asymptotic normality of TSLS (Appendix 12.3)

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- $E(q_i) = 0$ (by instrument exogeneity).
- q_i is i.i.d. with variance $\sigma_q^2 = \text{var}[(Z_i - \mu_Z) u_i]$.
- By the **CLT**: $\frac{\sqrt{n} \bar{q}}{\sigma_q} \xrightarrow{d} N(0, 1)$.

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The **denominator** converges in probability:

$$\frac{1}{n} \sum_{i=1}^n (Z_i - \bar{Z})(X_i - \bar{X}) \xrightarrow{p} \text{cov}(Z_i, X_i) \neq 0$$

By **Slutsky's theorem**, the ratio is asymptotically normal.

The large-sample distribution of TSLS

Theorem (Large-Sample Distribution)

Under the IV regression assumptions:

$$\widehat{\beta}_1^{TSLS} \overset{a}{\sim} N\left(\beta_1, \sigma_{\widehat{\beta}_1^{TSLS}}^2\right)$$

where

$$\sigma_{\widehat{\beta}_1^{TSLS}}^2 = \frac{1}{n} \cdot \frac{\text{var}[(Z_i - \mu_Z)u_i]}{[\text{cov}(Z_i, X_i)]^2} \quad (12.8)$$

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Key observations:

- Variance decreases at rate $1/n$ (as expected).
- Variance is **larger** when $\text{cov}(Z_i, X_i)$ is small $\Rightarrow$ weak instruments inflate variance.
- When $Z = X$ and $\text{cov}(Z, u) = 0$: reduces to OLS variance.

Statistical inference with TSLS

Because $\hat{\beta}_1^{TSLS}$ is asymptotically normal:

- **t-statistic:** $t = \frac{\hat{\beta}_1^{TSLS} - \beta_{1,0}}{SE(\hat{\beta}_1^{TSLS})}$
- **95% confidence interval:** $\hat{\beta}_1^{TSLS} \pm 1.96 \cdot SE(\hat{\beta}_1^{TSLS})$

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Two practical warnings

1. **Use TSLS standard errors**, not manual second-stage OLS standard errors (the latter are incorrect because they ignore first-stage estimation uncertainty).
2. **Use heteroskedasticity-robust standard errors** – exactly as with OLS.

Philip Wright's problem: supply and demand for butter

Philip Wright (1928) wanted to estimate the demand elasticity for butter:

$$\ln(Q_i^{butter}) = \beta_0 + \beta_1 \ln(P_i^{butter}) + u_i$$

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- Price and quantity are *jointly* determined by supply and demand.
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- OLS on the scatter of (P, Q) points estimates a meaningless hybrid.

Wright's insight: Find a variable that shifts *supply only* (not demand).

- When only supply shifts, the equilibrium traces out the **demand curve**.
- Instrument: **dairy-region rainfall**.

Why rainfall works as an instrument

Relevance: Below-average rainfall impairs grazing $\Rightarrow$ reduces butter supply $\Rightarrow$ raises equilibrium price.

$$\text{corr}(\text{Rainfall}, \ln P^{\text{butter}}) \neq 0 \checkmark$$

Exogeneity: Dairy-region rainfall has no direct effect on consumer *demand* for butter.

$$\text{corr}(\text{Rainfall}, u_i) = 0 \checkmark$$

Supply–demand diagram intuition

- Both curves shift: scatter of equilibria $\Rightarrow$ can't identify either curve.
- Only supply shifts: equilibria lie on the stable demand curve $\Rightarrow$ slope = demand elasticity.

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Historical note

IV regression was invented by Philip G. Wright and his son Sewall Wright (a geneticist) in *The Tariff on Animal and Vegetable Oils* (1928). The intellectual approach was anticipated by John Snow's 1855 cholera study, which used water company source as an instrument for exposure to contaminated water.

Application: demand elasticity for cigarettes

Policy question: How much must cigarette taxes rise to reduce consumption by 20%?

Answer depends on the **price elasticity of demand**:

$$\ln(Q_i^{cig}) = \beta_0 + \beta_1 \ln(P_i^{cig}) + u_i$$

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First stage (48 U.S. states, 1995):

$$\widehat{\ln(P_i^{cig})} = \underset{(0.03)}{4.62} + \underset{(0.005)}{0.031} \text{SalesTax}_i, \quad R^2 = 0.47$$

TSLS (second stage):

$$\widehat{\ln(Q_i^{cig})} = \underset{(1.53)}{9.72} - \underset{(0.32)}{1.08} \ln(P_i^{cig})$$

Demand is elastic: a 1% price increase reduces consumption by 1.08%.

The general model (Key Concept 12.1)

General IV Regression Model

$$Y_i = \beta_0 + \underbrace{\beta_1 X_{1i} + \cdots + \beta_k X_{ki}}_{\text{endogenous regressors}} + \underbrace{\beta_{k+1} W_{1i} + \cdots + \beta_{k+r} W_{ri}}_{\text{included exogenous/control variables}} + u_i \quad (12.12)$$

Four types of variables:

1. Y_i : dependent variable
2. $X_{1i}, \dots, X_{ki}$: k **endogenous** regressors (correlated with u_i)
3. $W_{1i}, \dots, W_{ri}$: r **exogenous** regressors or control variables
4. $Z_{1i}, \dots, Z_{mi}$: m **instrumental variables** (excluded from the structural equation)

Identification: counting instruments vs. endogenous regressors

The relationship between the number of instruments (m) and endogenous regressors (k) determines identification:

Case	Condition	Implication
Exactly identified	$m = k$	Can estimate; unique TSLS solution
Overidentified	$m > k$	Can estimate; can test overidentifying restrictions
Underidentified	$m < k$	Cannot estimate; not enough instruments

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Order condition (necessary)

IV regression requires at least as many instruments as endogenous regressors:
 $m \geq k$.

TSLS in the general model (Key Concept 12.2)

Two Stage Least Squares – General Case

First stage: For each endogenous regressor X_{ji} ($j = 1, \dots, k$), run the OLS regression:

$$X_{ji} = \pi_0 + \pi_1 Z_{1i} + \dots + \pi_m Z_{mi} + \pi_{m+1} W_{1i} + \dots + \pi_{m+r} W_{ri} + v_{ji}$$

Obtain predicted values $\hat{X}_{1i}, \dots, \hat{X}_{ki}$.

Second stage: Regress Y_i on $\hat{X}_{1i}, \dots, \hat{X}_{ki}, W_{1i}, \dots, W_{ri}$ using OLS.

The resulting estimators are $\hat{\beta}_0^{TSLS}, \hat{\beta}_1^{TSLS}, \dots, \hat{\beta}_{k+r}^{TSLS}$.

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The resulting estimators are $\hat{\beta}_0^{TSLS}, \hat{\beta}_1^{TSLS}, \dots, \hat{\beta}_{k+r}^{TSLS}$.

Key points:

- Each first-stage regression includes **all** instruments and **all** exogenous regressors.
- In practice, both stages are computed automatically by TSLS commands.

General conditions for valid instruments (Key Concept 12.3)

A set of instruments $Z_{1i}, \dots, Z_{mi}$ is valid if:

1. Instrument Relevance

Let $\hat{X}_{ji}^*$ be the predicted value from the *population* regression of X_{ji} on all Z 's and W 's. Then $(\hat{X}_{1i}^*, \dots, \hat{X}_{ki}^*, W_{1i}, \dots, W_{ri}, 1)$ are **not perfectly multicollinear**.

When $k = 1$: at least one Z has a nonzero coefficient in the population regression of X on the Z 's and W 's.

2. Instrument Exogeneity

$$\text{corr}(Z_{1i}, u_i) = 0, \quad \dots, \quad \text{corr}(Z_{mi}, u_i) = 0$$

All instruments must be uncorrelated with the error term.

IV regression assumptions (Key Concept 12.4)

The Four IV Regression Assumptions

1. $E(u_i | W_{1i}, \dots, W_{ri}) = 0$.
(Exogenous variables/controls are uncorrelated with u in the conditional mean sense.)
2. $(X_{1i}, \dots, X_{ki}, W_{1i}, \dots, W_{ri}, Z_{1i}, \dots, Z_{mi}, Y_i)$ are **i.i.d.** draws.
(Satisfied under simple random sampling.)
3. **Large outliers are unlikely:** all X 's, W 's, Z 's, and Y have nonzero finite fourth moments.
4. The two conditions for **valid instruments** (relevance + exogeneity) hold.

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3. **Large outliers are unlikely:** all X 's, W 's, Z 's, and Y have nonzero finite fourth moments.
4. The two conditions for **valid instruments** (relevance + exogeneity) hold.

Under these assumptions, $\hat{\beta}^{TSLS}$ is **consistent** and **asymptotically normal**.

The role of control variables (W 's) in IV regression

Why include exogenous variables W in the IV regression?

1. **Making instruments exogenous:** If Z is correlated with an omitted variable, but that variable is observable, include it as W . This breaks the correlation between Z and u .

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Cigarette demand example

Sales tax might correlate with state income, which affects demand. Including $\ln(\text{Inc}_i)$ as W breaks this correlation, making SalesTax a more credibly exogenous instrument:

$$\widehat{\ln(Q_i^{cig})} = 9.43 - \frac{1.14}{(1.26)} \ln(P_i^{cig}) + \frac{0.21}{(0.31)} \ln(\text{Inc}_i)$$

Weak instruments: why they are dangerous

If instruments explain very little of the variation in X , they are called **weak instruments**.

Consequences of weak instruments:

1. TSLS is **biased toward OLS**, even in large samples.
2. The normal approximation to the sampling distribution is **poor** $\Rightarrow$ t -statistics and confidence intervals are unreliable.
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Extreme case: irrelevant instrument

When $\text{cov}(Z_i, X_i) = 0$, the denominator $s_{ZX} \xrightarrow{P} 0$. The large-sample distribution of TSLS becomes the **ratio of two normal random variables** (not normal!), centered on the probability limit of OLS.

Weak instruments: the formal argument

From Eq. (12.20): $\hat{\beta}_1^{TSLS} = \beta_1 + \bar{q}/\bar{r}$ where:

$$\bar{q} = \frac{1}{n} \sum (Z_i - \bar{Z}) u_i \xrightarrow{d} N(0, \sigma_q^2/n)$$

$$\bar{r} = \frac{1}{n} \sum (Z_i - \bar{Z})(X_i - \bar{X})$$

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When instrument is irrelevant: $E(\bar{r}) = 0 \Rightarrow$ CLT applies to *both*
numerator and denominator:

$$\hat{\beta}_1^{TSLS} - \beta_1 \approx \frac{\sigma_q}{\sigma_r} \cdot \frac{\bar{q}/\sigma_{\bar{q}}}{\bar{r}/\sigma_{\bar{r}}} = \frac{\sigma_q}{\sigma_r} \cdot \frac{N(0, 1)}{N(0, 1)}$$

The distribution is a **ratio of two correlated normals** – heavy-tailed, centered at the OLS probability limit, not at β_1 .

Rule of thumb: first-stage $F > 10$ (Key Concept 12.5)

Key Concept 12.5: Checking for Weak Instruments

The **first-stage F -statistic** tests $H_0 : \pi_1 = \dots = \pi_m = 0$ in the first-stage regression.

Rule of thumb: If $F < 10$, the instruments are weak.

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Motivation (Appendix 12.5): The approximate bias of TSLS is:

$$E(\hat{\beta}_1^{TSLS}) - \beta_1 \approx \frac{\beta_1^{OLS} - \beta_1}{E(F) - 1}$$

- If $E(F) = 10$: TSLS bias is $\approx 1/9 \approx 11\%$ of OLS bias. Acceptable.
- If $E(F) = 5$: TSLS bias is $\approx 1/4 = 25\%$ of OLS bias. Worrisome.

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Stock–Yogo test

Formal version: compare the first-stage F (homoskedasticity-only) to critical values from Stock & Yogo (2005). For a single endogenous regressor, these critical values range from 9.08 to 11.52, so $F > 10$ is a good approximation.

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If you have weak instruments, three options:

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3. **Use robust inference methods** (Appendix 12.5):
 - **Anderson–Rubin (1949) test:** Valid whether instruments are strong, weak, or even irrelevant. Tests $H_0 : \beta_1 = \beta_{1,0}$ by regressing $Y_i - \beta_{1,0}X_i$ on Z 's and W 's and testing joint significance of Z 's.
 - **CLR test** (Moreira, 2003): Conditional likelihood ratio test with data-dependent critical values. More powerful than Anderson–Rubin; preferred when $k = 1$.

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Key point

With weak instruments, CLR or Anderson–Rubin *confidence sets* are preferable to point estimation. The TSLS point estimate itself may be badly biased.

Testing instrument exogeneity

Can we test whether instruments are exogenous?

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Important limitation

The J -test can never confirm that *all* instruments are exogenous. It tests consistency across instruments under the maintained assumption that at least k instruments are valid. Expert judgment is always required.

The J -statistic (Key Concept 12.6)

Overidentifying Restrictions Test

1. Estimate the IV regression by TSLS. Compute residuals:

$$\hat{u}_i^{TSLS} = Y_i - \hat{\beta}_0^{TSLS} - \hat{\beta}_1^{TSLS} X_{1i} - \dots - \hat{\beta}_{k+r}^{TSLS} W_{ri}$$

2. Regress $\hat{u}_i^{TSLS}$ on all instruments $Z_{1i}, \dots, Z_{mi}$ and all exogenous regressors $W_{1i}, \dots, W_{ri}$. Compute the homoskedasticity-only F -statistic testing $\delta_1 = \dots = \delta_m = 0$.
3. The J -statistic is:

$$J = m \times F$$

Under H_0 (all instruments exogenous): $J \sim \chi_{m-k}^2$

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$$J = m \times F$$

Under H_0 (all instruments exogenous): $J \sim \chi_{m-k}^2$

- Reject if $J > \chi_{m-k, \alpha}^2$ (or if $p\text{-value} < \alpha$).
- When $m = k$: $J \equiv 0$ (no overidentifying restrictions to test).

Intuition behind the J -test

Suppose $k = 1$ (one endogenous regressor) and $m = 2$ (two instruments).

- Compute TSLS using instrument 1 alone: $\hat{\beta}_1^{(1)}$
- Compute TSLS using instrument 2 alone: $\hat{\beta}_1^{(2)}$

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If both instruments are exogenous:

$$\widehat{\beta}_1^{(1)} \xrightarrow{p} \beta_1, \quad \widehat{\beta}_1^{(2)} \xrightarrow{p} \beta_1$$

They should be *close* (differing only by sampling variation).

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They should be *close* (differing only by sampling variation).

If the two estimates are **very different**, at least one instrument is likely endogenous.

The J -test formalizes this comparison (without actually computing all separate estimators).

Cigarette demand: panel data specification

Problem: Cross-sectional IV may still have omitted variables (e.g., historical tobacco culture).

Solution: Use 10-year differences to eliminate **state fixed effects**:

$$\underbrace{\ln Q_{i,95} - \ln Q_{i,85}}_{\Delta \ln Q} = \beta_0 + \beta_1 \underbrace{(\ln P_{i,95} - \ln P_{i,85})}_{\Delta \ln P} + \beta_2 \underbrace{(\ln \text{Inc}_{i,95} - \ln \text{Inc}_{i,85})}_{\Delta \ln \text{Inc}} + u_i$$

Instruments: $\Delta \text{SalesTax}$ and ΔCigTax (10-year changes).

This estimates the **long-run** price elasticity (10-year adjustment period allows for behavioral changes in addictive consumption).

Results: Table 12.1

	(1)	(2)	(3)
$\Delta \ln P^{cig}$	-0.94 (0.21)	-1.34 (0.23)	-1.20 (0.20)
$\Delta \ln Inc$	0.53 (0.34)	0.43 (0.30)	0.46 (0.31)
Instrument(s)	SalesTax	CigTax	Both
First-stage F	33.7	107.2	88.6
J -test (p -value)	—	—	4.93 (0.026)

All first-stage F 's $\gg 10 \Rightarrow$ instruments are **not weak**.

Interpreting the results

1. ***J*-test rejects** ($p = 0.026$): At least one instrument is endogenous.

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2. The two instruments give **different estimates**:
 - SalesTax only: elasticity = -0.94
 - CigTax only: elasticity = -1.34
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Interpreting the results

1. ***J*-test rejects** ($p = 0.026$): At least one instrument is endogenous.
2. The two instruments give **different estimates**:
 - SalesTax only: elasticity = -0.94
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3. **Which instrument to trust?**
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4. **Preferred estimate**: -0.94 (column 1, SalesTax instrument).

Interpretation: A 1% price increase reduces consumption by 0.94% in the long run. Cigarette demand is substantially elastic at the 10-year horizon.

Two approaches to finding instruments

Approach 1: Economic Theory

Use economic models to identify variables that shift one equation but not another.

- Wright's weather instruments for supply/demand.
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In both cases, **expert judgment is essential**. Finding good instruments is the hardest – and most important – part of IV analysis.

Example 1: Settler mortality → institutions → development

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- **Instrument:** Historical settler mortality rates among European colonizers.
- **Relevance:** High mortality $\Rightarrow$ “extractive” colonies $\Rightarrow$ weak institutions persisting to today. Settler mortality explains 27% of current institutional quality.
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Result: TSLS estimate of institutional effect is **twice the OLS estimate**, suggesting large simultaneous causality bias in OLS. Africa is poorer because of worse institutions, not geography.

Example 2: Random birth-date fluctuations → class size

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- **Instrument:** Deviations of potential kindergarten enrollment from long-term trend (random “spikes” from birth-date fluctuations).
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Result: TSLS estimates suggest the effect of class size on test scores is **small and statistically insignificant** – contrast with OLS which shows a significant effect (likely biased).

Example 3: Distance to hospital → treatment → survival

Question: Does cardiac catheterization prolong life for heart attack patients?

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McClellan, McNeil, Newhouse (1994):

- **Instrument:** Difference between distance to nearest catheterization hospital and distance to nearest hospital of any type.
- **Relevance:** Patients closer to catheterization hospitals more likely to receive treatment.
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Empirical check: distance uncorrelated with observable health measures.

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Result: TSLS finds **small, possibly zero** treatment effect. OLS shows large positive effect – likely biased by selection of healthier patients.

Caveat: IV estimates the effect for *compliers* (patients whose treatment is affected by distance), not the average treatment effect.

What happens when instruments are invalid?

Case 1: Irrelevant instrument ($\text{cov}(Z, X) = 0$)

- Distribution of TSLS is **ratio of two normals** (not normal).
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Case 2: Endogenous instrument ($\text{cov}(Z, u) \neq 0$)

TSLS is **inconsistent**:

$$\hat{\beta}_1^{TSLS} \xrightarrow{p} \beta_1 + \frac{\text{cov}(Z_i, u_i)}{\text{cov}(Z_i, X_i)} \neq \beta_1$$

The inconsistency depends on the ratio of instrument endogeneity to instrument relevance. Even a small violation of exogeneity matters if the instrument is weak.

Bias of TSLS with weak instruments

(Appendix 12.5)

Approximate bias formula

When there are m instruments:

$$E(\hat{\beta}_1^{TSLS}) - \beta_1 \approx \frac{\overbrace{\beta_1^{OLS} - \beta_1}^{\text{OLS bias}}}{E(F) - 1}$$

where $E(F)$ is the expected value of the first-stage F -statistic.

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$E(F)$	TSLS bias / OLS bias
100	1.0%
20	5.3%
10	11.1%
5	25.0%
2	100%

Anderson–Rubin test and confidence sets

When instruments may be weak, TSLS confidence intervals are unreliable.
Use instead:

Anderson–Rubin (1949) Test of $H_0 : \beta_1 = \beta_{1,0}$

1. Construct $Y_i^* = Y_i - \beta_{1,0}X_i$.
2. Regress Y_i^* on all W 's and Z 's.
3. Compute the F -statistic testing that all coefficients on Z 's are zero.

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regardless of instrument strength.

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Caveat

Anderson–Rubin confidence sets can be empty, unbounded, or disconnected – reflecting the genuine difficulty of identification with weak instruments.

CLR test (Moreira, 2003)

Conditional Likelihood Ratio Test

- Like a standard likelihood ratio test, but with **data-dependent critical values** that adjust for instrument strength.
- Valid whether instruments are strong, weak, or even irrelevant.
- More **powerful** than Anderson–Rubin when instruments are weak.
- Equivalent to the TSLS t -test when instruments are strong.

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Practical recommendation: When $k = 1$ and instruments might be weak, report both the standard TSLS confidence interval and the CLR confidence set. If they differ substantially, the instruments are likely weak.

Key takeaways

1. IV regression estimates causal coefficients when regressors are correlated with the error term.
2. **Endogenous** = correlated with u ; **exogenous** = uncorrelated with u .
3. A valid instrument must be: (1) **relevant** ($\text{corr}(Z, X) \neq 0$) and (2) **exogenous** ($\text{corr}(Z, u) = 0$).
4. IV regression requires $m \geq k$ (at least as many instruments as endogenous regressors).
5. TSLS works in two stages: (1) regress X on Z 's and W 's; (2) regress Y on $\hat{X}$'s and W 's.
6. **Weak instruments** bias TSLS toward OLS and invalidate standard inference. Check: first-stage $F > 10$.
7. **Endogenous instruments** make TSLS inconsistent. If overidentified, use the J -test. Otherwise, rely on expert judgment.